

Creating Curve-Based Garments with Custom Weave Patterns

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Figure 1: Weave structure images and close-up final render of woven garments. ©2023 Disney

ABSTRACT

In Walt Disney Animation Studios' "Strange World", the handmade quality of the costumes was a creative need for representing the technological limits of the setting, Avalonia. We expanded and standardized our curve-based fabric authoring workflows to provide simplified artistic controls, handle complex woven patterning, and support common finishing techniques like hems and stitches.

ACM Reference Format:

Dan Lipson and Jose Velasquez. 2023. Creating Curve-Based Garments with Custom Weave Patterns. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595443>

1 INTRODUCTION

Walt Disney Animation Studios' "Strange World" expanded on the studio's techniques for authoring fabric with accurately woven and knit digital threads. Curve-based garment workflows were first introduced on "Encanto". Look development artists constructed the fabric of Isabela's full dress from curves in Houdini, and Disney's XGen toolset was expanded to support curve-based cloth in our pipeline [Velasquez et al. 2022]. In this approach, each CV is given custom attributes with the face number and intrinsic faceUV coordinates of the nearest point on the garment mesh. Curves can be

bound to the garment in world space, or relative to cloth construction UVs using our cloth panel system. Using the per-CV binding information, the curves are deformed relative to the garment mesh as it is posed/simulated. The underlying garment mesh is hidden, leaving a fabric made entirely of curves that works through the rest of the pipeline and creates a rich material response that is not possible to emulate with surfacing shading alone.

2 WEAVE PATTERN GENERATION IN SUBSTANCE DESIGNER AND HOUDINI

While our technical setup was standardized with the work on "Encanto", there was no consistent workflow for artists to generate digital fabric swatches. To standardize this approach, we looked at traditional weaving to better understand the standards of real-world fabric creation. We found that weave patterns are often represented as black and white diagrams to indicate the structure of the horizontal weft threads and vertical warp threads. For our digital loom, a simple template in Adobe Substance Designer takes a black and white weave diagram as input and outputs an RGB image used to drive swatch curve geometry creation. Simple controls were exposed for artists to tune the curve profile and overall variation. The red, green, and blue channels of the output image are respectively used for the height of the weft threads, the height of the warp threads, and as a thread variation map (Figure 2).

In Houdini, curves are aligned to the template image and displaced based on the color channels. From there, multiple curves can be twisted along the threads to create a multi-ply yarn quality. The swatch is tiled into a larger sheet of fabric and cut into the shape of the garment construction panels. Finally, the per-CV attributes are computed and the curves are ready to plug into Disney's XGen and the rest of the pipeline.

As opposed to manually constructing weave patterns, or building a library of modeled patterns, authoring the pattern swatches with

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SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595443>

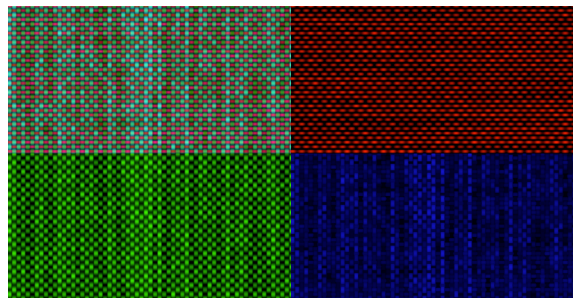


Figure 2: RGB texture used to drive swatch creation ©Disney

Substance Designer makes the workflow much more consistent and accessible. We can save and catalog these template patterns as textures alongside the rest of our texture library for easy sharing and reuse. Because the weave images share a standardized format, artists can swap between template images for notes or adjustments. The swatch size is only limited by texture resolution with our default template set to 192x192 because it is divisible by 2, 4, 6, 8, 12, and 16. This size is large enough to capture complex patterns as well as natural variation in the weave, helping avoid the uniformity of tiling small exemplar weave structures, while retaining enough pixel resolution to capture the height profile of each thread. To build new patterns, artists can rely on existing tools in Substance Designer or simply draw patterns by hand. Substance Designer is also used for creating mattes that line up with the source weave template, which helps when assigning different color or material properties to different threads, as seen in Figure 3.

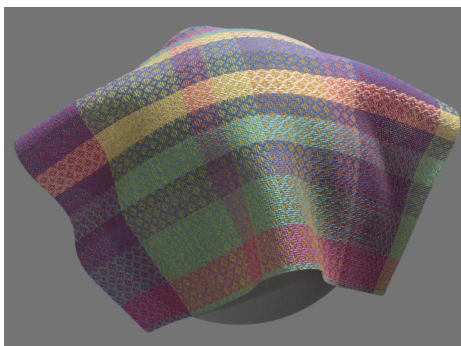


Figure 3: Woven fabric sample with a unique, custom weave pattern and authored per-thread color. ©Disney

3 KNITS, STITCHES, HEMS, AND SHADING

While woven patterns are based on grids which are easy to emulate digitally, knit fabrics can have much more complex structures. For Ethan’s knit jacket and his friend Azimuth’s seed stitch beanie, swatches were modeled by hand and converted to curves rather than using a procedural template.

In addition to creating the fabric, the final look also often needs stitching, hems, seams, and color. For stitches, artists can draw curves on the panels and procedurally convert them into a single

running stitch or whip stitch. Simple controls allow artists to line up scale and depth against the curve-based fabric and dial in a subtle distortion to add a hand-made irregularity.

For texturing and surfacing, custom RGB point attribute support was added to Disney’s XGen curve inputs. These attributes can be passed into our shaders and accessed just like any other texture variable. Attributes are painted on the garment mesh using our in-house paint package, Paint3D, and are projected onto the fabric curves in Houdini. We used the same projection techniques for micro wrinkles and seam puckering displacement which gets captured in the binding information.

Unlike physical garments, we do not tailor with accurate seam tolerances. For finished edges, like the bottom of a shirt or sleeve, we simply bent the cloth panel in Houdini to create a hem.



Figure 4: Before and after comparison of rim catching geometry intercepting light and filling silhouette. ©2023 Disney

4 GAPS AND TRANSLUCENCY

For rendering and simulation efficiency, we often make character geometry beneath fabric invisible. This practice is revealed by more accurate weaves and knits which reproduce the gaps between threads found in real fabric and a stronger sense of translucency in the material response. This problem is exacerbated when garments stretch since curve widths do not stretch with the weave structure. To fix this problem, we no longer make interior, camera-facing surfaces invisible. This sliver of geometry resolves issues with gaps and translucency by matching the silhouette, filling gaps, and intercepting rim lights, while avoiding collision with the front layer of fabric (Figure 4).

5 CONCLUSION

Our updated paradigm for creating swatches of woven patterns and the standardization of garment finishing techniques allowed us to author a total of 13 full garments across the film’s cast. This basic foundation was shared and reused by multiple look development artists and was able to expand to meet the unique challenges of our studio workflow and the artistic goals of the production. For future work we are exploring new ways to standardize artist control with respect to traditional weaving methods, such as control for weaving tension, improving the representation of yarns and threads in terms of structure and material response, and more robust options for representing real-world garment sewing processes like seam tolerances, curved hems, and other complex techniques.

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